# Pavithra Venkatachalapathy

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## Research Profile

Final-year Ph.D. candidate in Applied Mathematics specializing in computational fracture modeling and neural-network methods for differential equations. My doctoral research develops peridynamic simulations of bone fracture under blunt force impact. My broader work includes feedforward neural-network solvers, coding and writing contributions to physics-informed Kolmogorov–Arnold network studies, and color-image experiments for diffusion-based super-resolution. I am interested in hybrid numerical and machine-learning methods for scientific computing.

## Education

**Doctor of Philosophy in Applied Mathematics** *August 2022 – Present*
Texas Tech University, Lubbock, Texas, USA
GPA: 3.8/4.0
Expected completion: August 2027
Advisor: Dr. Victoria Howle
Dissertation: A Peridynamic Mathematical Model for Simulating Fracture Propagation in Cortical Bone Under Blunt Force Impact

**Master of Science in Applied and Computational Mathematics** *January 2020 – May 2022*
Texas A&M University–Corpus Christi, Texas, USA
GPA: 3.93/4.0
Advisor: Dr. Mallikarjunaiah S. Muddamallappa
Thesis: A feed-forward neural network framework for the solutions of linear and nonlinear ordinary differential equations

**Bachelor of Science in Civil Engineering** *August 2014 – June 2018*
Visvesvaraya Technological University, Karnataka, India
GPA: 3.95/4.0

## Research Interests

Computational fracture mechanics; peridynamics; scientific machine learning; numerical methods for differential equations; physics-informed neural networks; image reconstruction.

## Projects

- **Peridynamic Fracture Modeling** — Develop computational workflows and Peridigm simulations to investigate bone fracture under blunt force impact. Prepare CT-derived geometry and simulation inputs and compare preliminary simulated damage with experimental observations supplied by collaborators. *Peridigm, Python, ParaView.*
- **Neural-Network Solvers for Differential Equations** — Carried out research, implementation, numerical experiments, and manuscript writing for two coauthored journal papers on feedforward neural-network methods for nonlinear and singular ordinary differential equations. *Python, neural networks.*
- **KAN-Based Physics-Informed Solvers** — Contributed code and manuscript writing to collaborative studies of the Thomas–Fermi equation and higher-order boundary value problems using physics-informed Kolmogorov–Arnold networks. *Python, PyTorch.*
- **Diffusion-Based Image Super-Resolution** — Contributed code for color-image super-resolution experiments in a collaborative study of perceptually regularized diffusion models. *Python, PyTorch.*

## Publications

### Published Journal Articles

- S. M. Mallikarjunaiah and Pavithra Venkatachalapathy. *hp*-adaptive finite element simulation of a static anti-plane shear crack in a nonlinear strain-limiting elastic solid. *Finite Elements in Analysis and Design*, article 104520 (2026). doi:10.1016/j.finel.2026.104520.
- Pavithra Venkatachalapathy and S. M. Mallikarjunaiah. A deep learning neural network framework for solving singular nonlinear ordinary differential equations. *International Journal of Applied and Computational Mathematics*, 9, article 68 (2023). doi:10.1007/s40819-023-01563-x.
- Pavithra Venkatachalapathy and S. M. Mallikarjunaiah. A feedforward neural network framework for approximating the solutions to nonlinear ordinary differential equations. *Neural Computing and Applications*, 35, 1661–1673 (2023; published online October 2022). doi:10.1007/s00521-022-07855-5.

### Manuscripts Under Review

- Chuxiangbo Wang, Pavithra Venkatachalapathy, Ying Liang, Min Wang, Jing Qin, Yifei Lou, and Weihong Guo. Perceptually Regularized Diffusion Model for Image Super-Resolution. *(Under Review)* arXiv:2609.02016.
- Chandana Pati, B. Veena S.N. Rao, Pavithra Venkatachalapathy and S. M. Mallikarjunaiah. A Kolmogorov–Arnold Network Based Physics-Informed Approach for Solving the Thomas–Fermi Equation. *(Under Review)*
- Chandana Pati, B. Veena S.N. Rao, Pavithra Venkatachalapathy and S. M. Mallikarjunaiah. Physics-Informed Kolmogorov–Arnold Networks to Approximate Solutions to High-Order Boundary Value Problems. *(Under Review)*
- K. Nandini, Pavithra Venkatachalapathy, S. M. Mallikarjunaiah, Pramod Kumar Naik, T. G. Gangadhar, and Rupam Bhaduri. A Robust Deep Learning Framework for the Early Detection of Skin Cancer: Overcoming Data Imbalance to Achieve High-Precision Diagnostics. *(Under Review)*

### Manuscript in Preparation

- Pavithra Venkatachalapathy and Victoria Howle. Toward a Peridynamic Model for Simulating Blunt Force Fracture Patterns in Bone. *(Manuscript in preparation)*

## Talks, Presentations, and Conferences

### Presentations

- **Upcoming poster:** "A Hybrid Kolmogorov–Arnold Network and Finite Element Framework for Temporally Quasilinear Stress-Based Wave Equations." SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, Utah, November 16–20, 2026.
- **Poster presentation:** "Deep Learning Neural Network Frameworks for Solving Singular and Nonlinear Ordinary Differential Equations. Research at the Interface of Applied Mathematics and Machine Learning, CBMS Conference, University of Houston, Houston, Texas, December 8–12, 2025.
- **Poster presentation:** Spring Student Research Symposium, Texas A&M University–Corpus Christi, April 8, 2022. Best Poster Presentation Award.
- **Three Minute Thesis Competition:** Texas A&M University–Corpus Christi, April 7, 2022.

### Conference and Workshop Participation

- Research Collaboration Workshop in Science of Data and Mathematics (WiSDM), University of North Carolina at Chapel Hill, August 4–8, 2025.
- 18th U.S. National Congress on Computational Mechanics (USNCCM18), Chicago, Illinois, July 20–24, 2025.
- 5th and 6th Coastal Bend Mathematics and Statistics Conferences, 2021 and 2022.
- MSGSO 9th Annual Student Research Forum, Texas A&M University–Corpus Christi, April 30, 2021.

## Professional Experience

**Data Science using Machine Learning and Python – Internship** *2024*

*Dyashin LLC*

- *Completed a data science internship applying Python and machine learning techniques to real-world datasets.*

## Awards

- Graduate Student Research Support Award, Texas Tech University, 2026.
- SIAM Student Travel Award, 2026.
- Thomas McLaughlin Grad Math Scholarship, Texas Tech University, 2023.
- Best Poster Presentation at the 2022 Spring Student Research Symposium at Texas A&M University–Corpus Christi, April 08, 2022.
- College of S&E GR Scholarship, Texas A&M University–Corpus Christi, October 2021.
- TCRF Student Research Awards, Texas A&M University–Corpus Christi, September 2021.
- President's International Excellence Award, Texas A&M University–Corpus Christi, August 2020 and August 2021.
- Summer 2020 Graduate Scholarship, Texas A&M University–Corpus Christi, January 2020.

## Teaching Experience

**Graduate Part-Time Instructor** *August 2022 – Present*

*Department of Mathematics and Statistics, Texas Tech University*

- *Independently taught undergraduate mathematics courses, including Contemporary Mathematics (MATH 1300), Introductory Mathematical Analysis I (MATH 1330), and Analytical Geometry (MATH 1350).*

**Teaching Faculty** *May – July 2026*

*TexPREP-Lubbock*

- *Designed, developed, and delivered the curricula for Math Logic (Prep 1) and Statistics and Probability (Prep 3).*

**Independent Instructor** *January 2022 – May 2022*

*Department of Mathematics and Statistics, Texas A&M University–Corpus Christi*

- *Independently taught Elementary Statistics (MATH 1442) during the final semester of the master's program, following service as a Graduate Teaching Assistant.*

**Graduate Teaching Assistant** *January 2020 – May 2022*

*Department of Mathematics and Statistics, Texas A&M University–Corpus Christi*

- *Supported instruction for Elementary Statistics (MATH 1442), Statistics for Life (MATH 1442), Calculus I (MATH 2413), and Calculus II (MATH 2414), including lab instruction, grading, and office hours.*
- *Created recorded lab explanations and provided live online instruction to support student learning.*

## Leadership Experience

**Treasurer, SIAM-TTU Student Chapter** *July 2024 – July 2025*

*Society for Industrial and Applied Mathematics*

- *Prepared budgets and managed expenses. Organized monthly meetings to plan chapter activities and initiatives. Represented the chapter at university-wide events and conferences.*

## Technical Skills

- *Machine Learning & Scientific Computing*: Python, PyTorch, TensorFlow, Keras, NumPy, SciPy
- *Languages & Tools*: R, MATLAB, R-Spark, LaTeX, Git/GitHub  GitHub
- *Computational Mechanics & Visualization*: Peridigm, ParaView, Gmsh, meshio
- *Development Environments*: PyCharm, Visual Studio Code.
- *Data & Visualization*: Tableau, Power BI, MySQL

## Professional Membership

- American Mathematical Society.
- Society for Industrial and Applied Mathematics-Texas Tech Student Chapter.

## References

- **Dr. Victoria Howle (Academic Advisor)**
  Professor, Department of Mathematics and Statistics
  Texas Tech University
  Email: victoria.howle@ttu.edu
  Phone: +1 806-834-8770
- **Dr. Mallikarjunaiah S. Muddamallappa (Master's Thesis Advisor)**
  Associate Professor, Department of Mathematics and Statistics
  Texas A&M University–Corpus Christi
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